Supplementary Information

Crop-ranking reversal under downside risk and operational constraints

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Contents

1 Relationship to the supervisor draft

The immutable supervisor draft establishes the research question and innovation chain: suitability versus operational optimality; ranking reversal under uncertain price, yield and cost; joint downside risk and diversification failure; operational constraints; and the dependence of climate-information value on flexibility. The reconstruction retains that chain and the section architecture of introduction, literature, model, theory, numerical analysis, discussion and conclusion.

Several mathematical statements in the draft could not be retained as written. An undefined risk-adjusted margin gap did not imply an acreage "if and only if" result; KKT stationarity omitted operational multipliers and finite-scenario auxiliary variables; a scalar lower-tail coefficient did not globally order portfolio CVaR across copula families; a universal unique dependence threshold lacked monotonicity and crossing assumptions; and strict information–flexibility complementarity was not unconditional. The main paper replaces these statements with restricted exact theorems and reports null, disconnected and multiple-optimum cases.

The previous repository manuscript centered the proposition that an ordinal ranking does not identify a cardinal allocation. That statement is correct but does not by itself answer the supervisor's mechanism question. In the reconstructed manuscript it is treated as a boundary condition. The dominant argument is again the conditional reversal frontier under the full stochastic operational model.

2 Complete model notation

Symbol	Definition
$i = 1, \ldots, n$	crop index
ω	joint price–yield–cost state
s_i	external pre-decision agronomic or predictive score
x_i	planted area or normalized land share
$\widetilde{P}_i, \widetilde{Y}_i, \widetilde{C}_i$	stochastic price, yield and cost
$\widetilde{\pi}_i$	stochastic per-acre margin $\widetilde{P}_i \widetilde{Y}_i - \widetilde{C}_i$
μ_i	expected per-acre margin
$L(x)$	portfolio loss $-\widetilde{\pi}^\top x$
A	available land
B, c	working-capital limit and crop cash coefficients
R, r	rotation rows and limits
K, k	contractual-minimum rows and requirements
G, h	shared-capacity rows and limits
ℓ, u	crop-specific lower and upper area bounds
α	CVaR confidence level
κ	loss-CVaR ceiling
ρ	normalized risk-tolerance path index
$\mathcal{X}$	operational feasible set
$\mathcal{S}_\kappa$	complete optimal set under ceiling κ
σ	deterministic selector for one reported optimum
θ	parameter along a named joint-law family
ϕ	operational-flexibility or buffering parameter

3 Full finite-scenario linear programme

For scenario probabilities $w_m > 0$ summing to one, the implemented primal is

$$\max_{x,v,q} \quad \mu^\top x \tag{1}$$
$$\text{s.t.} \quad \mathbf{1}^\top x \le A, \tag{2}$$
$$c^\top x \le B, \tag{3}$$
$$Rx \le r, \qquad -Kx \le -k, \qquad Gx \le h, \tag{4}$$
$$\ell \le x \le u, \tag{5}$$
$$v + \frac{1}{1-\alpha}\sum_{m=1}^{M} w_m q_m \le \kappa, \tag{6}$$
$$-\pi_m^\top x - v - q_m \le 0, \qquad m = 1, \ldots, M, \tag{7}$$
$$q_m \ge 0, \qquad v \in \mathbb{R}. \tag{8}$$

There is no hidden penalty term and no post-solution repair of the optimized policy. Repairs apply only to heuristic benchmark policies so that they are compared inside the same operational set.

3.1 KKT equations

Let η be the CVaR-ceiling multiplier, t_m the multiplier for scenario excess row m, and z_m the multiplier on $q_m \ge 0$. Together with the operational multipliers defined in the main text, stationarity is

$$0 = -\mu + \gamma\mathbf{1} + \beta c + R^\top \rho - K^\top \chi + G^\top \lambda - a + b - \sum_m t_m \pi_m, \tag{9}$$
$$0 = \eta - \sum_m t_m, \tag{10}$$
$$0 = \frac{\eta w_m}{1-\alpha} - t_m - z_m, \qquad m = 1, \ldots, M. \tag{11}$$

All multipliers are nonnegative except the dual expression associated with the free variable v. Complementarity requires each multiplier times its primal slack to be zero. The scenario equations allow fractional mass at the VaR threshold and are valid for discrete scenario distributions.

4 Proofs

4.1 Proof of the restricted reversal theorem

Let $z = x_1$ and $x_2 = A - z$. Expected profit is

$$\mu_1 z + \mu_2(A - z) = \mu_2 A + (\mu_1 - \mu_2)z.$$

Because $\mu_1 > \mu_2$, this function is strictly increasing in z. Loss-CVaR is convex and continuous in the portfolio weights. Its sublevel set intersected with the compact operational interval I is therefore a compact interval $\mathcal{K}_{\theta,\kappa}$. A strictly increasing linear objective has the unique maximizer $\sup \mathcal{K}_{\theta,\kappa}$.

The pair reverses when $z < A - z$, equivalently $z < A/2$. Hence reversal holds if and only if the right endpoint of the entire risk-feasible interval lies below $A/2$, which is equivalent to no feasible point on $[A/2, \overline{z}]$. If risk is nondecreasing on that upper half, the minimum upper-half risk occurs at $A/2$, so the upper half is infeasible if and only if $r_\theta(A/2) > \kappa$. Existence of a feasible lower-half point ensures that this inequality indicates reversal rather than infeasibility.

If $\theta \mapsto r_\theta(A/2)$ is continuous, strictly increasing and crosses κ, the intermediate-value theorem gives a unique θ^*. If strictness is removed, an interval of thresholds can result. If monotonicity is removed, the reversal set can be disconnected. If crossing is removed, a threshold need not exist.

4.2 Proof of complete optimality

The finite-scenario problem is a bounded linear programme whenever $\mathcal{X}$ is nonempty and bounded. Linear-programming strong duality implies that primal and dual feasibility plus equality of the objective values are necessary and sufficient. Expanding equality of values across all inequality rows yields stationarity and complementarity. The equations above therefore form a complete certificate.

For the population convex formulation, write the Lagrangian using loss-CVaR as a convex function. Under Slater's condition or another convex constraint qualification, KKT conditions are necessary; convexity makes them sufficient. Pairwise subtraction of acreage stationarity cancels γ but leaves budget, rotation, contract, shared-resource and bound terms. No algebraic operation on this local equality recovers an acreage ordering without the global feasible geometry.

4.3 Proof of named-family ordering

For integrable random losses, CVaR is a law-invariant coherent risk measure and is monotone under convex order. Thus $L_{\theta_1}(x) \preceq_{\mathrm{cx}} L_{\theta_2}(x)$ implies the stated CVaR inequality at each x. Pointwise inequality gives containment of the two risk sublevel sets. Maximizing the same expected-profit objective over nested sets gives the optimal-value order. Coordinate monotonicity does not follow because the supporting face can change when the feasible set contracts.

4.4 Proofs for information and flexibility

Any no-information action defines a constant signal-contingent policy. If the constant policy is feasible, optimization after observing a signal cannot lower value. When one common action is posterior-optimal for every signal, the constant policy also attains the informed optimum, proving zero value. If unique posterior optima differ on events with positive probability and at least one difference is strict, expected improvement is positive.

Increasing differences and nested action sets yield monotone posterior actions under standard monotone comparative statics. Strictness on a positive- probability event makes the information gain larger at the higher acreage- flexibility level. This is the conditional complementarity result.

For buffering, at $\phi = 1$ all state margins equal $\bar{\pi}$; posterior optimization problems are identical and the common-action condition gives zero information value. If information value is positive at $\phi = 0$ and continuous along the contraction path, it must decline somewhere. Those intervals are substitutive. Total optimized value may still rise with ϕ.

5 Calibration and constraint table

Table 2: Registered Kansas calibration and primary allocation.

Crop	score	mean margin	margin s.d.	full-model share
		real 2024 US$ per acre		
Corn	0.748	199.8	94.9	0.330
Soybean	0.756	229.0	104.0	0.422
Winter wheat	0.894	138.3	49.6	0.248

Total land is 1.0, operating budget is US$330, planting-labour capacity is 0.82 and harvest-equipment capacity is 0.88. The full model uses $\alpha = 0.95$, $\rho = 0.5$, 512 optimization scenarios and seed 202607270034.

Table 3: Operational set. Coefficients use normalized land and capacity units.

Object	Corn	Soybean	Winter wheat
Lower bound	0.05	0.10	0.05
Upper bound	0.55	0.65	0.60
Rotation cap	0.50	—	—
Contract minimum	—	0.10	—
Planting-labour coefficient	1.00	0.72	0.58
Harvest-equipment coefficient	1.05	0.82	0.62

Table 4: Registered primary policy outcomes. A larger loss-CVaR is worse.

Policy	Corn	Soy	Wheat	Profit	loss-CVaR	risk feasible
Suitability proportional	0.312	0.315	0.373	187.4	-53.1	yes
Winner take all, repaired	0.300	0.100	0.600	166.6	-57.3	yes
Equal share	0.333	0.333	0.333	190.4	-51.4	yes
Expected profit	0.280	0.650	0.070	216.5	-34.5	no
Mean–variance	0.280	0.650	0.070	216.5	-34.5	no
Full loss-CVaR	0.330	0.422	0.248	198.5	-47.0	yes
Minimum-CVaR endpoint	0.220	0.180	0.600	169.0	-59.5	yes

6 Policy comparison

The identical primary expected-profit and mean–variance allocations are not a coding shortcut: under the registered variance penalty and constraints, both select the same boundary point. The diversification-failure test evaluates that point under the Student-t tail law and compares it with the full loss-CVaR solution.

7 Phase-diagram summary

Table 5: Family-specific reversal counts on each Kendall-τ path.

Family	τ	reversal	strong	multiple	first reversal	first strong
Clayton	0.00	9	5	2	0.2	0.6
Clayton	0.15	11	9	0	0.0	0.2
Clayton	0.30	10	7	0	0.0	0.4
Clayton	0.45	8	5	0	0.3	0.6
Clayton	0.60	9	3	0	0.2	0.8
Gaussian	0.00	11	10	0	0.0	0.1
Gaussian	0.15	10	5	0	0.1	0.6
Gaussian	0.30	10	8	0	0.1	0.3
Gaussian	0.45	8	2	0	0.3	0.9
Gaussian	0.60	10	5	0	0.1	0.6
Student-t	0.00	10	9	0	0.1	0.2
Student-t	0.15	10	9	0	0.1	0.2
Student-t	0.30	11	9	0	0.0	0.2
Student-t	0.45	8	7	0	0.3	0.4
Student-t	0.60	8	3	0	0.3	0.8

First-reversal and first-strong columns report the lowest risk-tolerance index on the registered grid.

Clayton at $\tau = 0.30$ is the only disconnected selected-reversal path. The two multiple-optimum cells remain in all counts and are resolved by separate possible, universal and selected classifications.

8 Robustness and uncertainty

Table 6: One-at-a-time robustness. "Selected" and "strong" are counts.

Dimension	cells	selected reversal	strong reversal
Confidence level	3	3	2
Constraint regime	5	5	3
Dependence family	3	3	2
Dependence parameter	5	5	4
Marginal law	3	3	3
Sample window	3	3	3
Scenario count	3	3	3
Random seed	3	3	2
Solver	3	3	3

The 64 historical bootstrap replications produce mean shares 0.277 for corn, 0.527 for soybean and 0.196 for winter wheat. Their 95% percentile intervals are [0.125, 0.368], [0.258, 0.650] and [0.070, 0.404]. Reversal probability is 0.969 with interval [0.892,0.996]. The first-reversal index mean is 0.066 with interval [0.000,0.598]. The broad boundary interval is consistent with an eight-year calibration and is not hidden by the large number of simulated scenarios.

9 Information–flexibility details

The state split contains four low and four high relative-yield-advantage years. Signal accuracy takes six values and flexibility takes six values on each path, producing 72 cells. The region counts are 25 strict-complementarity, 18 substitution, 18 zero and 11 weak or boundary. Every signal-contingent problem is feasible and the minimum computed value of information differs from zero only at floating-point scale.

At perfect signal accuracy, post-signal acreage reallocation raises information value monotonically from zero at no adjustable acreage to $ 4.655 at full adjustment. Under shock buffering, the corresponding information value declines toward zero as the state margins are contracted to a common vector. This comparison is about interaction; buffering can still increase the no-information optimized value.

10 External panel and identification boundary

Table 7: Top-rank disagreement in 248 state-years.

Ranking definition	estimate	cluster 95% low	cluster 95% high
Relative yield	0.887	0.802	0.954
Standardized revenue	0.613	0.447	0.767
Operating margin	0.625	0.552	0.693
Total-cost margin	0.782	0.701	0.857

The leakage-free relative-yield specification uses 651 crop-transition rows. Its prior-leader effect is -0.0042 with interval [-0.0123,0.0056]. The estimand is descriptive because the panel does not observe farm objectives, local prices or costs, credit, rotation history, contracts, risk preferences or received forecasts. State acreage is never substituted for an optimizer output.

11 Data lineage and reproducibility

The reconstruction begins with registered raw USDA NASS, USDA ERS and BLS files. The canonical state-crop panel is joined to national prices, costs and CPI-U through explicit crop and year keys. The Issue #34 script filters the frozen Kansas calibration years, constructs the genuine score and real margins, solves all policies, phase cells, robustness and bootstrap replications, and writes a machine-readable manifest.

The required commands are:

```
uv sync
uv run python scripts/run_issue34_reconstruction.py
(cd reconstruction/issue34/outputs && \
  shasum -a 256 -c SHA256SUMS.txt)
uv run python scripts/render_issue34_manuscript_numbers.py
uv run python scripts/make_issue34_figures.py
uv run pytest -q
```

The plotting script reads only registered output tables. Each figure is exported as editable SVG and PDF, 300-dpi PNG and 600-dpi TIFF. The manuscript and Supplementary Information compile from the repository root. The release archive contains the source registry, data dictionary, analysis manifest, checksums, tests, source data and rendered documents.

12 Limitations

1. The Kansas calibration has eight effective annual observations. It cannot estimate extreme-tail dependence reliably.
2. State yield combined with national price and cost is not a farm-level profit series.
3. Copula parameters and shared-capacity limits are registered stress values, not causal estimates.
4. The model is single-period and linear. It omits fixed charges, endogenous prices, spatial spillovers and explicit multi-year rotation state.
5. Historical bootstrap replications quantify sensitivity to the short calibration record but do not create independent empirical observations.
6. The information state is a transparent annual proxy, not a deployed forecast product.
7. The external panel is aggregate and descriptive; it cannot identify the full model's active mechanism or welfare.
8. Author order, affiliations and contribution wording require final confirmation before journal submission.